Comparative Analysis of Variational Quantum Methods for Hamiltonian Eigenvalue Problems

VQE, VQITE, and TTITE

quarkonium-simulations project

March 2025

Abstract

We compare three quantum-computing approaches to ground-state energy estimation implemented in the `quarksim` codebase, each drawn from a recent paper: (1) the Variational Quantum Eigensolver (VQE) of Gallimore & Liao, (2) Variational Quantum Imaginary-Time Evolution (VQITE) of Woloshyn, and (3) Trotter–Taylor Imaginary-Time Evolution (TTITE) of Yi *et al.* We focus on the *methods* rather than the specific physical systems, identifying where the approaches agree, differ, and what each can and cannot do. This document is intended as a planning reference for an apples-to-apples comparison of all three methods on the same Hamiltonian.

Contents

1 Overview **3**

2 Method 1: Variational Quantum Eigensolver (VQE) **3**
- 2.1 Principle . . . 3
- 2.2 Algorithm . . . 3
- 2.3 Ansatz: Unitary Coupled Cluster . . . 4
- 2.4 Excited States via Orthogonalization . . . 4
- 2.5 Error Mitigation: Zero-Noise Extrapolation . . . 4

3 Method 2: Variational Quantum Imaginary-Time Evolution (VQITE) **4**
- 3.1 Principle . . . 4
- 3.2 Algorithm: McLachlan's Variational Principle . . . 5
- 3.3 Ansatz . . . 5
- 3.4 Computing Gradients and the Metric Tensor . . . 5
- 3.5 Excited States via the Penalty Method . . . 6

4 Method 3: Trotter–Taylor Imaginary-Time Evolution (TTITE) **6**
- 4.1 Principle . . . 6
- 4.2 Step 1: Trotter Decomposition . . . 6
- 4.3 Step 2: Taylor Expansion . . . 6
- 4.4 Step 3: LCU Circuit with Ancilla Postselection . . . 7
- 4.5 Success Probability . . . 7
- 4.6 Excited States . . . 7

5 Detailed Comparison **8**
5.1 Convergence Mechanisms . . . 8
5.2 Sources of Error . . . 8
5.3 Circuit Resources . . . 8
5.4 Ansatz Dependence . . . 9
5.5 Noise Resilience . . . 9
5.6 Scalability . . . 9

6 Encoding Strategies **9**
6.1 Jordan–Wigner Encoding (Gallimore) . . . 10
6.2 Direct Encoding (Woloshyn, Yi *et al.*) . . . 10

7 Theoretical Quality and Guarantees **10**
7.1 Exactness in the Noiseless Limit . . . 10
7.2 Convergence Rates . . . 10
7.3 Summary of Theoretical Guarantees . . . 11

8 Strengths and Limitations Summary **11**
8.1 VQE . . . 11
8.2 VQITE . . . 11
8.3 TTITE . . . 12

9 Implications for the Apples-to-Apples Comparison **12**
9.1 Choosing the Test Problem . . . 12
9.2 Metrics for Comparison . . . 12
9.3 What to Watch For . . . 12

10 Conclusion **12**

1 Overview

All three methods solve the same abstract problem:

The Eigenvalue Problem

Given a Hamiltonian H acting on n qubits, find the ground-state energy E_0 and (optionally) the ground-state wavefunction $|\psi_0\rangle$.

The Hamiltonian is assumed to be expressible as a sum of Pauli products:

$$H = \sum_{i=1}^{m} c_i\, h_i, \qquad h_i \in \{I, X, Y, Z\}^{\otimes n} \tag{1}$$

where each h_i satisfies $h_i^2 = I$.

The three methods differ in *how* they extract E_0 from H. Table 1 provides a high-level comparison; the following sections elaborate.

	VQE (Gallimore)	**VQITE** (Woloshyn)	**TTITE** (Yi *et al.*)
Family	Variational	Variational	Direct ITE
Core idea	Minimize $\langle H\rangle$ via classical optimizer	Follow imaginary-time gradient on parameter manifold	Implement $e^{-H\tau}$ via Trotter + Taylor + LCU
Optimizer	Classical (COBYLA)	Natural gradient ODE	None (deterministic)
Ansatz	Required (UCC)	Required (R_y+CNOT)	Not required
Ancilla qubits	0	0	1 per Trotter step
Non-unitary?	No — pure unitary circuit	No — unitary approximation	Yes — postselection
Excited states	Orthogonalization	Penalty method	Not addressed
Error mitigation	ZNE	Readout correction	Not addressed

Table 1: High-level comparison of the three methods.

2 Method 1: Variational Quantum Eigensolver (VQE)

2.1 Principle

VQE exploits the **Rayleigh–Ritz variational principle**: for any normalized trial state $\left|\psi(\vec{\theta})\right\rangle$,

$$\boxed{E(\vec{\theta}) \;\equiv\; \left\langle \psi(\vec{\theta}) \middle| H \middle| \psi(\vec{\theta}) \right\rangle \;\geq\; E_0} \tag{2}$$

Minimizing $E(\vec{\theta})$ over $\vec{\theta}$ yields an upper bound on the true ground-state energy.

2.2 Algorithm

1. **Choose an ansatz** $U(\vec{\theta})$ — a parameterized quantum circuit.
2. **Prepare** $\left|\psi(\vec{\theta})\right\rangle = U(\vec{\theta})\,|0\rangle$.
3. **Measure** each Pauli term $\langle h_i\rangle$ separately; combine: $E(\vec{\theta}) = \sum_i c_i \langle h_i\rangle$.
4. **Update** $\vec{\theta}$ via a classical optimizer (e.g. COBYLA, SPSA, L-BFGS).
5. **Repeat** steps 2–4 until convergence.

The quantum computer acts as an *energy oracle*: given $\vec{\theta}$, it returns $E(\vec{\theta})$. All optimization logic runs classically.

2.3 Ansatz: Unitary Coupled Cluster

The Gallimore implementation uses a **Unitary Coupled Cluster (UCC)** ansatz specific to the single-particle sector:

$$U(\theta, \varphi) = \exp\{\theta(a_1^\dagger a_0 - a_0^\dagger a_1) + \varphi(a_2^\dagger a_0 - a_0^\dagger a_2)\} \tag{3}$$

Reparameterizing with $\alpha = \sqrt{\theta^2 + \varphi^2}$ and $\sin\beta = \theta/\alpha$ gives the most general single-particle state:

$$|\psi(\alpha, \beta)\rangle = \cos\alpha\ |0\rangle + \sin\alpha \sin\beta\ |1\rangle + \sin\alpha\cos\beta\ |2\rangle \tag{4}$$

parameterized by just **2 real numbers**. The circuit realization uses RY rotations and CNOT gates (depth 5, 8 gates total).

Ansatz expressibility

The UCC ansatz is *exactly* expressive for the single-particle Hilbert space: it can reach any state in $\text{span}\{|0\rangle, |1\rangle, |2\rangle\}$. This eliminates "ansatz error" for this problem but is a special feature of the single-particle sector — for multi-particle or larger systems, the ansatz may not span the full physical subspace.

2.4 Excited States via Orthogonalization

After finding the ground state $|\psi_0\rangle$, the first excited state is obtained by minimizing $E(\vec{\theta})$ subject to $\left\langle \psi_0 | \psi(\vec{\theta}) \middle| \psi_0 | \psi(\vec{\theta}) \right\rangle = 0$. This constraint reduces the parameter space from 2 to 1 dimension. The second excited state is then uniquely determined as the vector orthogonal to both.

Importantly, the paper proves that errors in excited-state estimates are bounded by the ground-state error: $|\delta_k| \leq \delta_0$ for all k.

2.5 Error Mitigation: Zero-Noise Extrapolation

On noisy hardware, VQE uses **zero-noise extrapolation (ZNE)**. Under a global depolarizing channel, each traceless Pauli expectation value scales as:

$$\langle h_i \rangle(\lambda) = A_i \cdot r^\lambda \tag{5}$$

where λ is a noise amplification factor controlled by **circuit folding**: replacing U with $U(U^\dagger U)^n$ gives $\lambda = 2n + 1$. Measuring at several λ values and fitting the exponential decay yields the noiseless estimate at $\lambda = 0$.

An improved model adds a constant offset to account for measurement-basis noise:

$$\langle h_i \rangle(\lambda) = A_i \cdot r^\lambda + c_i \tag{6}$$

3 Method 2: Variational Quantum Imaginary-Time Evolution (VQITE)

3.1 Principle

Imaginary-time evolution (ITE) provides a guaranteed path to the ground state. Expanding $|\psi(0)\rangle = \sum_k \alpha_k |E_k\rangle$ in the eigenbasis of H:

$$e^{-H\tau} |\psi(0)\rangle = \sum_k e^{-E_k \tau}\ \alpha_k\ |E_k\rangle \xrightarrow{\tau \to \infty} e^{-E_0 \tau}\ \alpha_0\ |E_0\rangle \tag{7}$$

Every component decays exponentially, and the ground state (smallest E_k) survives longest. The rate of convergence is controlled by the **spectral gap** $\Delta = E_1 - E_0$: after time τ, the excited-state contamination is suppressed by $e^{-\Delta\tau}$.

However, $e^{-H\tau}$ is **non-unitary** and cannot be directly implemented as a quantum gate. VQITE approximates ITE within a variational manifold.

3.2 Algorithm: McLachlan's Variational Principle

Parameterize the state as $\left|\psi(\vec{\theta})\right\rangle$ and seek parameter updates that best approximate ITE. McLachlan's principle leads to:

$$\boxed{\sum_j A_{ij}\,\dot{\theta}_j = C_i} \tag{8}$$

where:

$$C_i = -\frac{\partial E}{\partial \theta_i} \qquad \text{(energy gradient)} \tag{9}$$

$$A_{ij} = \mathrm{Re}\left[\frac{\partial \langle\psi|}{\partial\theta_i}\frac{\partial|\psi\rangle}{\partial\theta_j} - \frac{\partial\langle\psi|}{\partial\theta_i}|\psi\rangle\langle\psi|\frac{\partial|\psi\rangle}{\partial\theta_j}\right] \qquad \text{(quantum metric tensor / Fubini–Study metric)} \tag{10}$$

Parameters are updated via forward Euler:

$$\vec{\theta}(\tau+\Delta\tau) = \vec{\theta}(\tau) + \Delta\tau\,\dot{\vec{\theta}} \tag{11}$$

where $\Delta\tau$ acts as a learning rate.

Connection to natural gradient descent

The update rule (8) is equivalent to **natural gradient descent** on the energy landscape: $\dot{\vec{\theta}} = A^{-1}C$. The metric tensor A accounts for the curvature of the parameter manifold, preventing the optimizer from taking excessively large steps in directions where the state changes rapidly. This is why VQITE typically converges faster and more reliably than VQE with a first-order optimizer.

3.3 Ansatz

The Woloshyn implementation uses a simple **hardware-efficient ansatz** on 2 qubits:

$$\begin{array}{lcccccc} \mathrm{q}_0: & RY(\theta_0) & - & \bullet & - & & \\ \mathrm{q}_1: & RY(\theta_1) & - & \oplus & - & RY(\theta_2) & \end{array} \tag{12}$$

With 3 parameters, this circuit can represent any real superposition of $2^2 = 4$ basis states (3 free amplitudes after normalization).

3.4 Computing Gradients and the Metric Tensor

Energy gradient C_i: computed via the **parameter-shift rule** for R_Y gates:

$$\frac{\partial E}{\partial\theta_k} = \frac{E(\theta_k+\pi/2) - E(\theta_k-\pi/2)}{2} \tag{13}$$

This is exact (not a finite-difference approximation) because $R_Y(\theta) = e^{-i\theta Y/2}$ generates sinusoidal dependence. Cost: $2p$ circuit evaluations per step (p = number of parameters).

Metric tensor A_{ij}: computed via finite differences on the overlap function. Cost: $O(p^2)$ circuit evaluations per step. Regularization $A \to A + \lambda I$ ($\lambda \sim 10^{-4}$) is applied to handle near-singular cases.

3.5 Excited States via the Penalty Method

To find the k-th excited state, the effective energy is modified:

$$E_{\text{eff}}(\vec{\theta}) = \left\langle \psi(\vec{\theta}) \middle| H \middle| \psi(\vec{\theta}) \right\rangle + \alpha \sum_{j<k} \left| \left\langle \phi_j | \psi(\vec{\theta}) \middle| \phi_j | \psi(\vec{\theta}) \right\rangle \right|^2 \tag{14}$$

where $\{|\phi_j\rangle\}_{j<k}$ are previously determined lower-lying states and $\alpha \gg \Delta$ is a penalty coefficient. The penalty term raises the energy of any state that overlaps with known eigenstates, effectively "pushing" the variational state toward the k-th eigenstate.

The method is general (works for any ansatz and any number of excited states) but requires choosing α large enough, and errors accumulate sequentially.

4 Method 3: Trotter–Taylor Imaginary-Time Evolution (TTITE)

4.1 Principle

TTITE implements imaginary-time evolution *directly* on a quantum computer, without any variational ansatz. The non-unitary operator $e^{-H\tau}$ is decomposed into a product of operators that can each be realized via an ancilla qubit and postselection.

The algorithm has three layers of approximation:

1. **Trotter decomposition**: factorize $e^{-H\Delta\tau}$ into a product of single-term exponentials.
2. **Taylor expansion**: truncate each exponential to finite order.
3. **LCU postselection**: implement each truncated operator as a linear combination of unitaries using an ancilla qubit.

4.2 Step 1: Trotter Decomposition

For $H = \sum_{i=1}^{m} c_i h_i$, the first-order Trotter formula gives:

$$e^{-H\Delta\tau} \approx \prod_{i=1}^{m} e^{-c_i h_i \Delta\tau} + O(\Delta\tau^2) \tag{15}$$

The full evolution over time τ is obtained by repeating $L = \tau/\Delta\tau$ such **Trotter segments**:

$$e^{-H\tau} \approx \left(\prod_{i=1}^{m} e^{-c_i h_i \Delta\tau} \right)^L \tag{16}$$

The Trotter error is $O(\Delta\tau)$ per segment and $O(\Delta\tau)$ overall (since $L\Delta\tau = \tau$ is fixed). This is empirically the dominant error source.

4.3 Step 2: Taylor Expansion

Since each h_i is a Pauli product ($h_i^2 = I$), the Taylor series of $e^{-c_i h_i \Delta\tau}$ collapses to just two terms:

$$e^{-c_i h_i \Delta\tau} \approx T_i^{(R)}(\Delta\tau) = \underbrace{\sum_{\substack{j=0 \\ j \text{ even}}}^{R} \frac{(-c_i\Delta\tau)^j}{j!}}_{\alpha_i} I + \underbrace{\sum_{\substack{j=0 \\ j \text{ odd}}}^{R} \frac{(-c_i\Delta\tau)^j}{j!}}_{\beta_i} h_i \tag{17}$$

In the limit $R \to \infty$: $\alpha_i \to \cosh(c_i\Delta\tau)$, $\beta_i \to -\sinh(c_i\Delta\tau)$. In practice, $R = 5$ suffices for small $\Delta\tau$.

Key insight

Each Trotter step reduces to a **linear combination of two unitaries**: $T_i = \alpha\, I + \beta\, h_i$. This is the crucial simplification that makes quantum circuit implementation possible.

4.4 Step 3: LCU Circuit with Ancilla Postselection

The non-unitary operator $\alpha I + \beta h_i$ is implemented using one ancilla qubit:

1. **Encode**: Apply $R_Y(\theta)$ on the ancilla, with $\theta = 2\arccos\left(\alpha/\sqrt{\alpha^2+\beta^2}\right)$:

$$|0\rangle\,|\psi\rangle \;\longrightarrow\; \frac{1}{\sqrt{\alpha^2+\beta^2}}\big(\alpha\,|0\rangle + \beta\,|1\rangle\big)\,|\psi\rangle \tag{18}$$

2. **Controlled-h_i**: Apply h_i on the work register conditioned on the ancilla being $|1\rangle$:

$$\longrightarrow\; \frac{1}{\sqrt{\alpha^2+\beta^2}}\big(\alpha\,|0\rangle\,|\psi\rangle + \beta\,|1\rangle\,h_i\,|\psi\rangle\big) \tag{19}$$

3. **Hadamard** on the ancilla:

$$\longrightarrow\; \frac{1}{\sqrt{2(\alpha^2+\beta^2)}}\big[|0\rangle\,(\alpha I + \beta h_i)\,|\psi\rangle + |1\rangle\,(\alpha I - \beta h_i)\,|\psi\rangle\big] \tag{20}$$

4. **Measure**: If ancilla $= |0\rangle$, the work system is in state $(\alpha I + \beta h_i)\,|\psi\rangle$ (normalized).

4.5 Success Probability

The probability of the ancilla collapsing to $|0\rangle$ is:

$$P_s = \frac{\|(\alpha I + \beta h_i)\,|\psi\rangle\,\|^2}{2(\alpha^2+\beta^2)} \tag{21}$$

An improved circuit replaces the Hadamard with a tuned $R_Y^\dagger$ rotation, achieving:

$$P_s' = \frac{\|T_i\,|\psi\rangle\,\|^2}{(\alpha+\beta)^2} \;\geq\; P_s \tag{22}$$

The postselection bottleneck

Each Trotter step requires a successful ancilla measurement. For a full Trotter segment of m terms, the cumulative success probability is $P_{\text{seg}} = \prod_{i=1}^{m} P_{s,i}$. Over L segments:

$$P_{\text{total}} = \prod_{\ell=1}^{L}\prod_{i=1}^{m} P_{s,i}^{(\ell)} \tag{23}$$

If each step has $P_s \approx 0.9$ and there are $m = 5$ terms per segment and $L = 30$ segments, then $P_{\text{total}} \approx 0.9^{150} \approx 10^{-7}$. This exponential decay in success probability is the main practical limitation of TTITE.

4.6 Excited States

The paper does not address excited-state computation. In principle, one could use the **quantum phase estimation** (QPE) approach or the **orthogonal filtering** technique, but these are not part of the TTITE framework as presented.

5 Detailed Comparison

5.1 Convergence Mechanisms

The three methods reach the ground state through fundamentally different mechanisms:

VQE	A classical optimizer explores the energy landscape $E(\vec{\theta})$, seeking its global minimum. Convergence depends on the optimizer, the landscape geometry (barren plateaus, local minima), and the number of function evaluations.
VQITE	The parameters follow a deterministic ODE (Eq. 8) that approximates imaginary-time evolution within the variational manifold. Convergence is guaranteed if: (a) the ansatz can represent the ground state, and (b) the metric tensor A is well-conditioned.
TTITE	Imaginary-time evolution is implemented directly; the ground state emerges by exponential suppression of excited components. Convergence rate: $\sim e^{-\Delta\tau}$ per unit imaginary time, where $\Delta = E_1 - E_0$ is the spectral gap.

5.2 Sources of Error

Error source	VQE	VQITE	TTITE
Ansatz expressibility	✓	✓	—
Classical optimizer (local minima)	✓	—	—
Barren plateaus	✓	✓*	—
Finite imaginary time	—	✓	✓
Trotter error	—	—	✓
Taylor truncation	—	—	✓
Postselection failure	—	—	✓
Shot noise	✓	✓	✓
Gate noise	✓	✓	✓
Metric tensor ill-conditioning	—	✓	—

Table 2: Error sources for each method. *Barren plateaus affect VQITE indirectly through the gradient C_i and metric tensor A_{ij}.

5.3 Circuit Resources

Resource	VQE	VQITE	TTITE
Work qubits	n	n	n
Ancilla qubits	0	0	1
Circuits per iteration	$O(m)$	$O(m \cdot p + p^2)$	$O(m \cdot L)$
Circuit depth	Ansatz depth	Ansatz depth	$O(n)$ per step
Classical optimizer calls	50–500	0	0
Total quantum cost	Optimizer-dependent	Deterministic	Deterministic

Table 3: Resource scaling. n: qubits, m: Pauli terms in H, p: number of variational parameters, L: Trotter segments.

VQE requires $O(m)$ circuit evaluations per optimizer call (one per Pauli term), with ∼50–500 optimizer iterations. Each circuit has depth equal to the ansatz depth.

VQITE requires $O(p^2 + mp)$ circuit evaluations per imaginary-time step: $O(p^2)$ for the metric tensor and $O(mp)$ for the gradient. With ∼30–50 steps, this is typically more expensive than VQE per step but converges more reliably.

TTITE requires $m \times L$ sequential circuit executions, each needing a successful ancilla postselection. The per-step circuit depth is $O(n)$ (controlled-Pauli gates), but the cumulative success probability decays exponentially with the total number of steps.

5.4 Ansatz Dependence

- **VQE and VQITE** both require an ansatz, and the quality of the result is bounded by the ansatz's expressibility. If the ground state lies outside the variational manifold, neither method can find it.
- **TTITE** is *ansatz-free*: it operates directly on the full 2^n-dimensional Hilbert space. The only requirement is that the initial state has nonzero overlap with the ground state.

This is TTITE's most significant theoretical advantage: it cannot suffer from ansatz error.

5.5 Noise Resilience

VQE is often described as "noise-resilient" because:

1. Short ansatz circuits accumulate less gate error.
2. The classical optimizer can partially adapt to systematic noise.
3. ZNE provides a systematic correction framework.

VQITE inherits similar noise properties (short circuits, finite shot tolerance) but has no built-in error mitigation beyond readout correction. The metric tensor computation is sensitive to noise because it involves second-order information.

TTITE is the *least* noise-resilient:

1. Deep circuits (many sequential Trotter steps) accumulate gate errors.
2. Postselection failure compounds across steps.
3. No variational "self-correction" mechanism.

5.6 Scalability

Aspect	VQE	VQITE	TTITE
Qubit count	Scales with encoding	Scales with encoding	Same + 1 ancilla
Parameter scaling	p grows with system; barren plateau risk $\sim 2^n$	Same as VQE	No parameters
Circuit depth	Shallow (ansatz-limited)	Shallow (ansatz-limited)	Deep ($\sim m \cdot L$)
Postselection	None	None	Exponential overhead
Demonstrated	2–3 qubits (charmonium)	2 qubits (charmonium)	Up to 12 qubits (Heisenberg chain)

Table 4: Scalability comparison.

6 Encoding Strategies

An orthogonal but important design choice is how the physical Hamiltonian is mapped to qubits. The three papers use two different strategies:

6.1 Jordan–Wigner Encoding (Gallimore)

The Hamiltonian is written in second-quantized form $H = \sum_{mn} h_{mn}\, a_m^\dagger a_n$, and fermionic operators are mapped to Pauli strings via the Jordan–Wigner transformation:

$$a_n^\dagger = \frac{1}{2}\left(\prod_{j<n} Z_j\right)(X_n - iY_n) \tag{24}$$

$$a_n = \frac{1}{2}\left(\prod_{j<n} Z_j\right)(X_n + iY_n) \tag{25}$$

For N orbitals, this uses N qubits but only the single-particle sector (N of 2^N states) is physical. The resulting Hamiltonian has $O(N^2)$ Pauli terms. For Gallimore's $N = 3$ system: 3 qubits, 10 Pauli terms, but only 3 of 8 computational basis states are physical.

Advantage: Systematic and general—applicable to any second-quantized Hamiltonian (molecules, lattice models, etc.).

Disadvantage: Wastes qubits. The unphysical states must be avoided or projected out.

6.2 Direct Encoding (Woloshyn, Yi *et al.*)

The N basis states are mapped directly to $\lceil \log_2 N \rceil$ qubits: $|k\rangle \leftrightarrow$ the binary representation of k. For Woloshyn's $N = 4$ system: 2 qubits, all 4 computational basis states are physical.

Advantage: Maximally qubit-efficient; the full Hilbert space is physical.

Disadvantage: Requires computing the Hamiltonian matrix in the chosen basis and decomposing it into Pauli strings (no general recipe like Jordan–Wigner).

Encoding is independent of the method

For the apples-to-apples comparison, all three methods should use the *same* encoding and the *same* Hamiltonian matrix. The direct encoding is more qubit-efficient and avoids unphysical states, making it the natural choice.

7 Theoretical Quality and Guarantees

7.1 Exactness in the Noiseless Limit

- **VQE**: Exact if and only if the ansatz can represent the ground state *and* the optimizer finds the global minimum. Both conditions are hard to guarantee in general.
- **VQITE**: Converges to the best approximation of the ground state within the variational manifold. The natural gradient prevents local minima trapping (but does not eliminate ansatz error).
- **TTITE**: In the limit $\Delta\tau \to 0$ (no Trotter error), $R \to \infty$ (no Taylor error), and $\tau \to \infty$ (full convergence), TTITE is *exact* regardless of system size. It requires no ansatz and has no local minima.

7.2 Convergence Rates

VQE: Optimizer-dependent. COBYLA typically requires $O(p^2)$ to $O(p^3)$ function evaluations. May get stuck in local minima for large p.

VQITE: The energy decreases monotonically (by construction) at a rate related to the spectral gap Δ. In the linear regime:

$$E(\tau) - E_0 \sim e^{-2\Delta\tau} \tag{26}$$

In practice, convergence in ∼20–50 steps with $\Delta\tau = 0.02$.

TTITE: Same exponential convergence as exact ITE, modulated by Trotter and Taylor errors:

$$|E_\tau - E_0| \leq C_1\, e^{-\Delta\tau} + C_2\, \Delta\tau + C_3\, (\Delta\tau)^R / R! \tag{27}$$

where C_1 is the initial contamination, C_2 captures Trotter error, and C_3 captures Taylor error. The Trotter error dominates in practice.

7.3 Summary of Theoretical Guarantees

Property	VQE	VQITE	TTITE
Variational upper bound	Yes	Yes	No*
Guaranteed convergence	No	Conditional†	Yes‡
Exact in principle	Ansatz-limited	Ansatz-limited	Yes
Exponential convergence rate	No	Yes	Yes
Robust to noise	Moderate	Moderate	Low

Table 5: *TTITE gives the correct energy but not a strict upper bound (Trotter error can go either direction). †Guaranteed if the ansatz is expressive enough and the metric tensor is non-singular. ‡Guaranteed if the initial state has nonzero overlap with the ground state and $\Delta\tau$ is small enough.

8 Strengths and Limitations Summary

8.1 VQE

Strengths:

- Shallowest circuits — most practical on near-term hardware.
- Flexible choice of optimizer and ansatz.
- ZNE provides systematic noise correction.
- Well-studied excited-state methods (orthogonalization, deflation).

Limitations:

- No convergence guarantee (local minima, barren plateaus).
- Ansatz design is an art: too simple ⇒ biased results; too expressive ⇒ barren plateaus.
- High measurement overhead for many Pauli terms.
- Optimizer performance degrades with system size.

8.2 VQITE

Strengths:

- Deterministic parameter updates (no optimizer hyperparameters).
- Natural gradient avoids many local-minimum issues.
- Guaranteed monotonic energy decrease.
- General penalty method for excited states.

Limitations:

- Still ansatz-limited (same expressibility constraint as VQE).
- $O(p^2)$ metric tensor evaluation is expensive for many parameters.
- Metric tensor can become ill-conditioned near critical points.
- More circuit evaluations per step than VQE.

8.3 TTITE

Strengths:

- Ansatz-free — exact in principle.
- Deterministic, no optimizer or hyperparameters (beyond $\Delta\tau$, R).
- Guaranteed convergence for any initial state with ground-state overlap.
- Scalable to many qubits in classical simulation.

Limitations:

- Postselection probability decays exponentially with circuit depth.
- Deep circuits make it very sensitive to gate noise.
- Trotter error is the dominant error source.
- No natural excited-state method.
- Each Trotter step requires resetting and re-measuring the ancilla.

9 Implications for the Apples-to-Apples Comparison

9.1 Choosing the Test Problem

For a fair comparison, the test Hamiltonian should:

1. Be small enough for exact diagonalization (ground truth).
2. Use a single encoding (direct, not Jordan–Wigner) so all methods see the same qubit Hamiltonian.
3. Include spin-dependent interactions (to match the Woloshyn problem and make it physically interesting).

The Woloshyn charmonium system (4×4 matrix, 2 qubits, 10 Pauli terms, three spin channels) is a natural candidate.

9.2 Metrics for Comparison

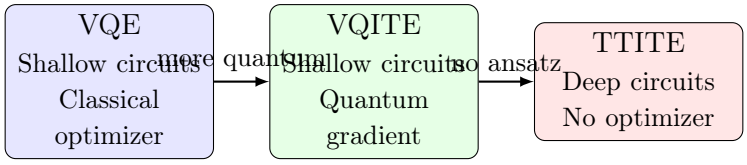
1. **Accuracy**: $|E_{\text{computed}} - E_{\text{exact}}|$ for ground and excited states.
2. **Convergence speed**: number of circuit evaluations (or wall-clock time) to reach a target accuracy.
3. **Noise sensitivity**: accuracy degradation as a function of simulated gate error rate.
4. **Circuit cost**: total number of CNOT gates and circuit depth.
5. **Scalability**: how metrics change if we increase the basis size (e.g., $N = 4 \to 8 \to 16$, requiring $2 \to 3 \to 4$ qubits).

9.3 What to Watch For

- **VQE**: does the optimizer consistently find the global minimum, or does it get trapped? How many random restarts are needed?
- **VQITE**: does the metric tensor stay well-conditioned? How sensitive is convergence to $\Delta\tau$ and the initial parameters?
- **TTITE**: what is the cumulative success probability after full convergence? How does Trotter error compare to the variational error of VQE/VQITE?
- **All methods**: how do they degrade under realistic noise models (depolarizing, amplitude damping, readout error)?

10 Conclusion

The three methods represent a spectrum of approaches to quantum eigenvalue problems:



VQE is the most practical for near-term noisy devices but offers the weakest theoretical guarantees. VQITE improves convergence reliability through natural gradient dynamics but retains the ansatz limitation. TTITE eliminates the ansatz entirely and is exact in principle, but its exponential postselection overhead limits practical applicability.

The upcoming apples-to-apples comparison will quantify these trade-offs on a concrete problem, providing empirical evidence for which method offers the best accuracy-per-quantum-resource on near-term hardware.